

Handwritten Digit Recognition System

Objective

This project uses Artificial Intelligence and Deep Learning to recognize handwritten digits from images using the MNIST dataset.

Technologies Used

- Python
- TensorFlow
- Keras
- CNN (Convolutional Neural Network)

Install Required Libraries

```
pip install tensorflow keras matplotlib
```

Python Code

```
from tensorflow.keras.datasets import mnist
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten

(x_train, y_train), (x_test, y_test) = mnist.load_data()

model = Sequential()
model.add(Flatten(input_shape=(28,28)))
model.add(Dense(128, activation='relu'))
model.add(Dense(10, activation='softmax'))

model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

model.fit(x_train, y_train, epochs=3)

loss, accuracy = model.evaluate(x_test, y_test)
print("Accuracy:", accuracy)
```

Features

- Recognizes handwritten digits
- Uses MNIST dataset
- Deep learning model training
- Accuracy evaluation

Concepts Used

- Deep Learning
- Neural Networks
- Image Processing

- CNN Model

Sample Output

```
Epoch 1/3  
Accuracy: 0.97
```

```
Model successfully recognized handwritten digits.
```

GitHub Description

This project is a Handwritten Digit Recognition System developed using Python, TensorFlow, and Deep Learning concepts. It recognizes handwritten numbers using the MNIST dataset.

LinkedIn Post

Successfully completed Handwritten Digit Recognition System project using Python, TensorFlow, and Deep Learning during my Artificial Intelligence Internship at SkillInfyTech.

Viva Questions

1. What is Deep Learning?
2. What is CNN?
3. What is the MNIST dataset?
4. Why is TensorFlow used?
5. What is model accuracy?

Conclusion

This project demonstrates how Deep Learning and Artificial Intelligence can be used for image recognition and classification tasks.